Lowering of vapour pressure

1. Vapour pressure of CCl_4 at 25^oC is $143mm$ of $Hg 0.5gm$ of a non-volatile solute (mol. wt. = 65) is dissolved in $100ml CCl_4$. Find the vapour pressure of the solution (Density of $CCl_4 = 1.58g/cm^2$)
(a) $141.43mm$ (b) $94.39mm$
(c) $199.34mm$ (d) $143.99mm$

2. For a solution of volatile liquids the partial vapour pressure of each component in solution is directly proportional to
(a) Molarity
(b) Mole fraction
(c) Molality
(d) Normality

3. "The relative lowering of the vapour pressure is equal to the mole fraction of the solute." This law is called
(a) Henry's law
(b) Raoult's law
(c) Ostwald's law
(d) Arrhenius's law

4. The relative lowering of vapour pressure produced by dissolving 71.5 *g* of a substance in 1000 *g* of water is 0.00713. The molecular weight of the substance will be
(a) 18.0 (b) 342
(c) 60 (d) 180

5. When mercuric iodide is added to the aqueous solution of potassium iodide, the
(a) Freezing point is raised
(b) Freezing point is lowered
(c) Freezing point does not change
(d) Boiling point does not change

6. Vapour pressure of a solution is
(a) Directly proportional to the mole fraction of the solvent
(b) Inversely proportional to the mole fraction of the solute
(c) Inversely proportional to the mole fraction of the solvent
(d) Directly proportional to the mole fraction of the solute

7. When a substance is dissolved in a solvent the vapour pressure of the solvent is decreased. This results in
(a) An increase in the b.p. of the solution
(b) A decrease in the b.p. of the solvent
(c) The solution having a higher freezing point than the solvent
(d) The solution having a lower osmotic pressure than the solvent

8. If P^o and P are the vapour pressure of a solvent and its solution respectively and N_1 and N_2 are the mole fractions of the solvent and solute respectively, then correct relation is
(a) $P = P^o N_1$
(b) $P = P^o N_2$
(c) $P^o = PN_2$
(d) $P = P^o (N_1/N_2)$

9. An aqueous solution of methanol in water has vapour pressure
(a) Equal to that of water
(b) Equal to that of methanol
(c) More than that of water
(d) Less than that of water

10. The pressure under which liquid and vapour can coexist at equilibrium is called the
(a) Limiting vapour pressure
(b) Real vapour pressure
(c) Normal vapour pressure
(d) Saturated vapour pressure

11. Which solution will show the maximum vapour pressure at 300 *K*
(a) 1 *M* $C \rightleftarrows_{12} H_{22}O_{11}$
(b) 1 *M* CH_3COOH
(c) 1 *M* $NaCl_2$
(d) 1 *M* $NaCl$

12. The relative lowering of the vapour pressure is equal to the ratio between the number of
(a) Solute moleules and solvent molecules
(b) Solute molecules and the total molecules in the solution
(c) Solvent molecules and the total molecules in the solution
(d) Solvent molecules and the total number of ions of the solute

13. $5cm^3$ of acetone is added to $100cm^3$ of water, the vapour pressure of water over the solution
(a) It will be equal to the vapour pressure of pure water
(b) It will be less than the vapour pressure of pure water
(c) It will be greater than the vapour pressure of pure water
(d) It will be very large

14. At 300 *K*, when a solute is added to a solvent its vapour pressure over the mercury reduces from 50 *mm* to 45 *mm*. The value of mole fraction of solute will be
(a) 0.005 (b) 0.010
(c) 0.100 (d) 0.900

15. A solution has a 1 : 4 mole ratio of pentane to hexane. The vapour

pressure of the pure hydrocarbons at 20°*C* are 440 *mmHg* for pentane and 120 *mmHg* for hexane. The mole fraction of pentane in the vapour phase would be

(a) 0.549 (b) 0.200
(c) 0.786 (d) 0.478

16. Benzene and toluene form nearly ideal solutions. At 20°*C*, the vapour pressure of benzene is 75 *torr* and that of toluene is 22 *torr*. The parial vapour pressure of benzene at 20°*C* for a solution containing 78*g* of benzene and 46*g* of toluene in *torr* is

(a) 50 (b) 25
(c) 37.5 (d) 53.5

17. The vapour pressure lowering caused by the addition of 100 *g* of sucrose(molecular mass = 342) to 1000 *g* of water if the vapour pressure of pure water at 25^oC is 23.8 *mm Hg*

(a) 1.25 *mm Hg*
(b) 0.125 *mm Hg*
(c) 1.15 *mm Hg*
(d) 00.12 *mm Hg*

18. Which of the following is incorrect

(a) Relative lowering of vapour pressure is independent
(b) The vapour pressure is a colligative property
(c) Vapour pressure of a solution is lower than the vapour pressure of the solvent
(d) The relative lowering of vapour pressure is directly propertional to the original pressure

19. Among the following substances the lowest vapour pressure is exerted by

(a) Water
(b) Mercury
(c) Kerosene
(d) Rectified spirit

20. According to Raoult's law the relative lowering of vapour pressure of a solution of volatile substance is equal to

(a) Mole fraction of the solvent
(b) Mole fraction of the solute
(c) Weight percentage of a solute
(d) Weight percentage of a solvent